

CAUSANTA: Causal Analysis Using Somatic And Neighborhood Tissue Architecture

Theoretical Framework & Simulation Engine Specification

1. Motivation and Scientific Context

Spatial transcriptomics and proteomics capture high-dimensional tissue organization, but current computational frameworks are limited to descriptive analysis. Existing methods do not distinguish causal drivers from downstream effects, nor do they predict tissue responses to molecular perturbations. These limitations stem from the observational nature of spatial data, which is heavily confounded by microenvironmental "neighborhood effects" that mask true causal drivers. Fundamental questions remain unanswered: *Which computational approaches can reliably identify causal relationships? How can we reconstruct tissue evolution from static data? How can we predict tissue responses to molecular perturbations?*

CAUSANTA addresses these gaps by treating somatic stochasticity as endogenous biological randomization, enabling causal and temporal inference from multimodal spatial data. The key innovation is to treat heritable cell-to-cell genomic variation as Somatic Instrumental Variables (SIVs) within a Structural Causal Model. This enables identification of causal effects from otherwise observational spatial data. Extrachromosomal DNA (ecDNA) segregation serves as an ideal validation system: its randomized inheritance acts as a natural experiment, and its genomic ground truth enables rigorous benchmarking. While ecDNA anchors validation, the framework generalizes to other forms of heritable somatic variation, enabling causal inference across diverse diseases and biological contexts.

The CAUSANTA simulation engine generates synthetic tissue sections with known ground-truth causal structure, enabling rigorous benchmarking of causal discovery algorithms. By simulating evolutionary trajectories within tissues — treating somatic variation as a biological clock — CAUSANTA supports inference of temporal progression, identification of lineage-specific drivers of clonal expansion, and *in silico* molecular intervention experiments that explore how specific perturbations alter the tissue microenvironment.

2. Spatial Domain and Coordinate System

2.1 Domain

CAUSANTA simulates a 2D tissue section representing a histological slice. The domain size is configurable up to **12 mm × 12 mm** (12,000 μm × 12,000 μm).

2.2 Coordinate Precision

The **minimal spatial unit is 1 μm** . All cell positions (x, y) are recorded in micrometer coordinates at integer precision. This provides sub-cellular coordinate resolution for accurate spatial registration with FISH, PhenoCycler, and H&E imaging data.

2.3 Environment Grid

Environment fields (O_2 , glucose, etc.) are computed on a **coarser regular grid** to maintain computational tractability. The environment grid resolution is a configurable parameter (`env_grid_um`, default 10 μm). For a 1 mm $\times$ 1 mm domain at 10 μm resolution, the environment grid is $100 \times 100 = 10,000$ voxels. For a 12 mm domain at 10 μm resolution, it is $1,200 \times 1,200 = 1,440,000$ voxels.

Cell agents exist in continuous (integer μm) space and sample the environment grid via bilinear interpolation at their position.

2.4 Time

The **time step is 1 hour**. All rate parameters are expressed per hour. Internal diffusion sub-stepping occurs within each 1-hour step as needed for numerical stability.

3. Cell Agent Model

3.1 Cell Types

Type ID	Cell Type	Typical Diameter (μm)	Role
0	Neuron	10-20	Parenchyma; post-mitotic; displaced/killed by tumor
1	Astrocyte	10-15 (soma)	Glial support; tiles in non-overlapping ~ 50 -100 μm domains; reactive astrogliosis at tumor margin
2	Oligodendrocyte	8-12	Myelinating glia; displaced by tumor; some gliomas have oligo-lineage features
3	Microglia	8-10	Resident immune; 5-10% of glia; activates and migrates toward tumor
4	Endothelial	10-15	Vasculature; O_2 /glucose source; angiogenesis
5	Pericyte	5-8	Wraps endothelial cells; blood-brain barrier integrity; vascular remodeling

Type ID	Cell Type	Typical Diameter (μm)	Role
6	Tumor (Glioma)	15-25	Proliferative, invasive, ecDNA-bearing
7	Recruited Immune	10-12	T cells, macrophages entering from vasculature
8	Necrotic	variable	Dead cell; releases DAMPs; occupies space until cleared

3.2 Cell State Vector

Every cell agent c carries a complete state record at each time step:

cell_id	int	Unique identifier (monotonically increasing)
parent_id	int	cell_id of mitotic parent (-1 for initial cells)
cell_type	int	Type code (0-8, see 3.1)
cell_type_name	string	Human-readable name
# Spatial		
x	int	X position in μm (1 μm precision)
y	int	Y position in μm
angle	float	Orientation angle in radians $[0, 2\pi)$
# Morphology		
shape_path	string	SVG path string defining cell outline (Vega mark path format)
size_scale	float	Scale factor applied to shape_path (1.0 = default for type)
# Proliferation		
cell_cycle_phase	string	{G0, G1, S, G2, M}
cycle_clock_hr	float	Hours elapsed in current phase
# Genomic / ecDNA		
ecDNA_count	int	Copy number of extrachromosomal DNA elements
ecDNA_cargo	string	Semicolon-delimited amplified genes (e.g., "EGFR;MYC")
# Lineage		
generation	int	Division count from founding ancestor
time_born_hr	float	Simulation hour when cell was created
# Local environment (sampled)		
O2_local	float	Local oxygen (mmHg) at cell position
glucose_local	float	Local glucose (mM) at cell position
# Derived state		
is_hypoxic	bool	$O_2 < \text{hypoxia threshold}$
is_quiescent	bool	In G0 (contact inhibited or resource-starved)
# Current rates (effective, after ecDNA modulation)		
migration_rate	float	Effective migration speed ($\mu\text{m/hr}$)
VEGF_secretion	float	Effective VEGF output (amol/hr)

3.3 Cell Shape Representation

Cell shapes are defined as **SVG path strings**, following the same convention as Vega's `path`

mark type. Each cell type has a default shape path normalized to a unit bounding box (roughly -1 to 1 in both axes). The `size_scale` factor scales the path uniformly, and `angle` rotates it.

Default shape paths by cell type:

```
json
{
  "Neuron": "M 0 -1 C 0.5 -0.5 0.3 0.2 0.8 1 L 0 0.6 L -0.8 1 C -0.3 0.2 -0.8 1",
  "Astrocyte": "M 0 -1 L 0.4 -0.3 L 1 -0.2 L 0.4 0.2 L 0.5 1 L 0 0.5 L -0.5 1",
  "Oligodendrocyte": "M 0 -0.8 A 0.8 0.8 0 1 1 0 0.8 A 0.8 0.8 0 1 1 0 -0.8 Z",
  "Microglia": "M 0 -0.6 L 0.3 -0.3 L 0.8 -0.5 L 0.5 0 L 0.8 0.5 L 0.3 0.3 L 0",
  "Endothelial": "M -1 -0.3 Q 0 -0.5 1 -0.3 L 1 0.3 Q 0 0.5 -1 0.3 Z",
  "Pericyte": "M -0.6 -0.4 Q 0 -0.6 0.6 -0.4 L 0.6 0.4 Q 0 0.6 -0.6 0.4 Z",
  "Tumor": "M 0 -0.9 C 0.6 -0.9 0.9 -0.4 0.9 0 C 0.9 0.5 0.5 0.9 0 0.9 C -",
  "RecruitedImmune": "M 0 -0.7 A 0.7 0.7 0 1 1 0 0.7 A 0.7 0.7 0 1 1 0 -0.7 Z",
  "Necrotic": "M -0.5 -0.5 L 0.5 -0.5 L 0.5 0.5 L -0.5 0.5 Z"
}
```

The rendering pipeline (Vega.js) transforms each cell's path by: scale by $\text{size_scale} \times (\text{diameter} / 2)$, rotate by `angle`, translate to `(x, y)`.

3.4 Cell Type Parameters

Each cell type τ carries a parameter set $\mathbf{P}_\tau$. All rates are **per hour** (the simulation time unit).

```

# Identity
type_id                int
type_name              string
shape_path             string    SVG path for default shape
color                 string    Hex color for visualization

# Geometry
diameter_mean_um      float    Mean cell body diameter (μm)
diameter_std_um       float    Standard deviation (μm)

# Proliferation
can_divide            bool      Whether this type can undergo mitosis
division_time_mean_hr float    Mean total cell cycle duration (hours)
division_time_std_hr  float    Stochastic variation (hours)
O2_prolif_threshold_mmHg float  Minimum O2 for proliferation
glucose_prolif_threshold_mM float Minimum glucose for proliferation
max_generations       int      Division limit (-1 = unlimited)

# Migration
migration_speed_um_hr float    Base migration speed (μm per hour)
migration_persistence_hr float  Directional memory time constant (hours)
chemotaxis_O2         float    Coefficient: bias toward O2 gradient
chemotaxis_VEGF       float    Coefficient: bias toward VEGF gradient
chemotaxis_chemokine  float    Coefficient: bias toward chemokine gradient
haptotaxis_ECM        float    Coefficient: bias along ECM gradient
contact_inhibited     bool      Whether migration stops at confluent
neighbors

# Death
O2_necrosis_threshold_mmHg float  O2 below which necrosis initiates
necrosis_delay_hr      float    Hours below threshold before death
apoptosis_rate_per_hr  float    Spontaneous death probability per hour
lysis_rate_per_hr      float    Rate at which necrotic cells are cleared

# Metabolism & Secretion
O2_consumption_amol_hr float    Oxygen consumption (attomoles per hour)
glucose_consumption_amol_hr float Glucose consumption (attomoles per hour)
VEGF_secretion_amol_hr float    VEGF production when hypoxic
lactate_production_amol_hr float  Lactate output (Warburg effect)

# State transition probabilities (per hour, conditional on trigger)
transition_rules       list      Each entry: {condition, target_type,
rate_per_hr}

```

```

# ecDNA (tumor cells only)
ecDNA_init_count      int      Starting ecDNA copy number
ecDNA_cargo           list     Gene names on ecDNA
ecDNA_segregation_p   float    Per-copy probability of going to daughter
(0.5 = random)
ecDNA_effect_on_division float   $\alpha$ : proliferation acceleration per
 $\log_2(\text{ecDNA})$ 
ecDNA_effect_on_VEGF  float     $\beta$ : VEGF secretion multiplier per ecDNA copy
ecDNA_effect_on_migration float   $\delta$ : migration speed multiplier per ecDNA
copy
ecDNA_effect_on_survival float   $\gamma$ : apoptosis resistance per ecDNA copy

```

Reference Values

Parameter	Neuron	Astrocyte	Oligo	Microglia	Endothelial	Pericyte
diameter (μm)	15 ± 3	12 ± 2	10 ± 2	9 ± 1	12 ± 2	6 ± 1
can_divide	no	yes	yes	yes	yes	yes
division_time (hr)	—	168 ± 24	168 ± 24	72 ± 12	60 ± 12	96 ± 12
migration ($\mu\text{m/hr}$)	0	3 ± 1	1 ± 0.5	30 ± 10	10 ± 3	5 ± 2
O ₂ consumption (amol/hr)	30000	12000	9000	6000	3000	2000
O ₂ prolif threshold (mmHg)	—	10	10	5	8	8
O ₂ necrosis threshold (mmHg)	2.5	2.5	2.5	1.0	2.5	2.5
VEGF secretion (amol/hr, hypoxic)	0	30	0	0	0	0
apoptosis rate (/hr)	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
color	#4477AA	#66CCEE	#228833	#EE6677	#AA3377	#BBBBBB

4. Environment Field Model

4.1 Substrates and Governing Equation

Each environment field E_k is defined on the environment grid (resolution env_grid_um) and evolves via a reaction-diffusion partial differential equation:

$$\partial E_k / \partial t = D_k \nabla^2 E_k - \lambda_k E_k + \text{sources}(x,t) - \text{sinks}(x,t)$$

where:

- D_k

 = diffusion coefficient ($\mu\text{m}^2/\text{hr}$)
- \lambda_k

 = natural decay rate (1/hr)
- sources

 = supply from vasculature and cell secretion
- sinks

 = cellular consumption

4.2 Primary Fields

Field	Symbol	Units	D_k ($\mu\text{m}^2/\text{hr}$)	\lambda_k (/hr)	Normal Value	Role
Oxygen	O ₂	mmHg	6.0×10^6	~0	38 mmHg (~5%)	Proliferation, hypoxia, necrosis, HIF
Glucose	Gluc	mM	2.4×10^6	~0	5.0 mM	Metabolic fuel; Warburg effect
VEGF	VEGF	nM	3.6×10^4	0.6	0 nM	Angiogenesis trigger
Lactate	Lac	mM	9.0×10^5	0.06	1.0 mM	pH proxy; immunosuppressive
ECM density	ECM	[0-1]	0 (static)	0	0.5	Migration scaffold; tumor remodeling
Vascular density	Vasc	[0-1]	0 (static)	0	varies	O ₂ /glucose source; immune entry

Note: Diffusion coefficients converted to $\mu\text{m}^2/\text{hr}$ from standard $\mu\text{m}^2/\text{min}$ values ($\times 60$).

4.3 Environment Grid Record

Each environment grid voxel at position (i, j) stores:

x_grid	int	Grid column index
y_grid	int	Grid row index
x_um	float	Physical X position (μm)
y_um	float	Physical Y position (μm)
O ₂	float	Oxygen partial pressure (mmHg) [0-60]
glucose	float	Glucose concentration (mM) [0-5.5]
VEGF	float	VEGF concentration (nM) [0-100]
lactate	float	Lactate concentration (mM) [0-40]
ECM_density	float	Extracellular matrix density [0-1]
vascular_density	float	Local vessel density [0-1]
pH	float	Derived: $7.4 - 0.02 \times \text{lactate}$
is_occupied	bool	Whether a cell occupies this voxel center
occupant_cell_id	int	cell_id of occupant (-1 if empty)
occupant_type	int	Cell type of occupant (-1 if empty)

4.4 Source and Sink Terms

Oxygen supply from vasculature:

$$\text{source}_{\text{O}_2}(\mathbf{x}) = q_{\text{O}_2} \times \text{vasc_density}(\mathbf{x}) \times (\text{O}_2_{\text{blood}} - \text{O}_2(\mathbf{x}))$$

where $\text{O}_2_{\text{blood}} = 60 \text{ mmHg}$, q_{O_2} is a transfer coefficient (hr^{-1}).

Oxygen consumption by cells:

$$\text{sink}_{\text{O}_2}(\mathbf{x}) = \sum_c \text{O}_2_{\text{consumption}_c} \times \delta(\mathbf{x} - \mathbf{x}_c) / V_{\text{voxel}}$$

where the sum runs over cells c located within the voxel, and $V_{\text{voxel}} = (\text{env_grid_um})^2$ is the voxel area.

VEGF secretion (hypoxia-dependent):

$$\text{source}_{\text{VEGF}}(\mathbf{x}) = \sum_c \text{VEGF_rate}_c \times H(\theta_{\text{hypoxia}} - \text{O}_2(\mathbf{x}_c)) \times \delta(\mathbf{x} - \mathbf{x}_c) / V_{\text{voxel}}$$

where H is the Heaviside function.

Glucose follows the same source/sink pattern as oxygen. **Lactate** is produced by tumor cells proportional to glucose consumption (Warburg effect).

4.5 Boundary Conditions

Dirichlet (fixed value) at domain edges, representing surrounding normal tissue:

- $\text{O}_2 = 38 \text{ mmHg}$

- Glucose = 5.0 mM
- VEGF = 0 nM
- Lactate = 1.0 mM

4.6 Numerical Diffusion Scheme

Implicit Locally One-Dimensional (LOD) operator splitting, solving each spatial direction via the Thomas algorithm (tridiagonal matrix solve):

For each substrate, within each 1-hour time step, diffusion is sub-stepped at `dt_diffusion` (configurable, default 6 seconds = 0.1 min):

1. X-sweep: solve tridiagonal system implicit in x for each row
2. Y-sweep: solve tridiagonal system implicit in y for each column
3. Apply source/sink terms as a separate operator step
4. Enforce boundary conditions
5. Clamp values to physical ranges

The implicit scheme is **unconditionally stable** — no CFL constraint on `dt_diffusion`, allowing large sub-steps even with oxygen's high diffusion coefficient.

5. Cell Behavioral Rules

All rates in this section are **per hour** unless otherwise noted.

5.1 Proliferation (Mitosis)

For each cell *c* whose type permits division, at each time step:

1. Resource gate:

```
can_proliferate = (O2_local > O2_prolif_threshold)
                  AND (glucose_local > glucose_prolif_threshold)
                  AND (generation < max_generations OR max_generations == -1)
                  AND (at least one adjacent position unoccupied)
```

2. Cell cycle progression:

Phase durations are fractions of the total division time *T* (drawn once per cycle from `Normal(mean, std)`):

- G1: $0.40 \times T$

- S: $0.30 \times T$
- G2: $0.15 \times T$
- M: $0.15 \times T$

`cycle_clock_hr` advances by 1 (the time step) each hour. Phase transitions occur when accumulated time exceeds phase duration. If resources drop below threshold during S/G2/M, the cell arrests (clock pauses, does not revert).

3. Division (M phase completion):

- Select a random unoccupied position adjacent to the parent (8-connected neighborhood)
- If no position available → cell enters G0 (contact inhibition), clock pauses
- If position available:
 - Create daughter cell with new `cell_id`, `parent_id` = parent's `cell_id`
 - Daughter placed at selected position
 - Both cells reset (`cycle_clock_hr = 0`), enter G1
 - `generation` incremented for both cells

ecDNA segregation (the SIV mechanism):

```
For parent with ecDNA_count = N:
    daughter_ecDNA = Binomial(N, p)    where p = ecDNA_segregation_p (default 0.5)
    parent_ecDNA   = N - daughter_ecDNA
```

This binomial partitioning is the core mechanism that generates inter-cell ecDNA heterogeneity — creating the "natural experiment" that enables causal inference. The randomness of segregation satisfies the independence assumption required for instrumental variable analysis.

ecDNA modulation of cell behavior:

```
division_time_effective = division_time_base / (1 + α × log2(1 + ecDNA_count))
VEGF_secretion_effective = VEGF_secretion_base × (1 + β × ecDNA_count)
migration_effective = migration_base × (1 + δ × ecDNA_count)
apoptosis_effective = apoptosis_base / (1 + γ × ecDNA_count)
```

These are the **ground-truth causal effects** that CAUSANTA embeds in the simulation. The causal discovery algorithms being benchmarked should recover these relationships from the simulated spatial data.

5.2 Migration

Each hour, motile cells compute a migration vector and move:

1. Direction:

$$\mathbf{v} = \mathbf{v}_{\text{persist}} + \chi_{\text{O}_2} \nabla \text{O}_2 + \chi_{\text{VEGF}} \nabla \text{VEGF} + \chi_{\text{chemo}} \nabla \text{chemokine} + \chi_{\text{ECM}} \nabla \text{ECM} + \mathbf{v}_{\text{noise}}$$

- $\mathbf{v}_{\text{persist}}$ = previous direction $\times \exp(-1/\text{persistence_time})$, providing directional memory
- Gradient terms: environment fields sampled at cell position, finite-difference gradient over neighboring voxels
- $\mathbf{v}_{\text{noise}}$ = uniform random angle perturbation

2. Displacement:

$$\begin{aligned}\Delta x &= \text{migration_speed} \times \cos(\theta_v) \times 1 \text{ hr} \\ \Delta y &= \text{migration_speed} \times \sin(\theta_v) \times 1 \text{ hr}\end{aligned}$$

Position updated: $(x \ += \ \text{round}(\Delta x), y \ += \ \text{round}(\Delta y))$ (integer μm).

3. Collision: If target position is within one cell diameter of another cell, the move is rejected (volume exclusion). The cell may attempt an alternate direction or stay.

Type-specific behaviors:

- **Neurons, Oligodendrocytes:** migration_speed = 0 (sessile)
- **Astrocytes:** Low motility; increased near tumor margin (reactive astrogliosis)
- **Microglia:** High motility; strong chemotaxis toward DAMPs from necrotic regions
- **Tumor:** Moderate motility; biased toward O_2 gradients, along ECM fibers; optional Go-or-Grow coupling: $\text{speed_effective} = \text{speed_base} \times (1 - \text{proliferation_activity})$
- **Recruited Immune:** Highest motility; chemotaxis toward tumor-secreted chemokines
- **Pericytes:** Low motility; constrained to remain adjacent to endothelial cells

5.3 Cell Death

Necrosis: When $\text{O}_2_{\text{local}} < \text{O}_2_{\text{necrosis_threshold}}$ for necrosis_delay_hr consecutive hours:

- $\text{cell_type} \rightarrow 8$ (Necrotic)

- Cell remains at position, blocks space, ceases metabolism
- Cleared stochastically: $P(\text{lysis}) = \text{lysis_rate} \times 1 \text{ hr}$ per time step

Apoptosis: At each step:

- $P(\text{apoptosis}) = \text{apoptosis_rate_per_hr} \times 1 \text{ hr}$
- Apoptotic cells removed from grid immediately

Immune-mediated kill: When a recruited immune cell or activated microglia is within kill_radius of a tumor cell:

- $P(\text{kill}) = \text{kill_rate} \times \text{activation_level} \times 1 \text{ hr}$

5.4 Cell Type Transitions

From	Trigger Condition	To	Rate (per hr)
Astrocyte	Adjacent to tumor cell	Reactive Astrocyte (subtype flag)	0.01
Microglia	Local chemokine > threshold	Activated Microglia (subtype flag)	0.05
Activated Microglia	Local TGF- β > threshold	Immunosuppressed Microglia	0.02
Endothelial	Local VEGF > angio_threshold	Sprouting Endothelial (subtype flag)	0.01
Any living cell	$O_2 < \text{necrosis threshold}$ (sustained)	Necrotic	deterministic
(vascular source)	Local chemokine > recruit_threshold	Recruited Immune (new cell spawned)	0.005

5.5 Angiogenesis

When VEGF exceeds $\text{angiogenesis_threshold_nM}$ at a position adjacent to an existing endothelial cell:

1. **Tip cell selection:** endothelial cell with highest local VEGF
2. **Sprouting:** tip cell extends one position per hour up the VEGF gradient; a new endothelial cell is placed at the vacated position (stalk cell)
3. **Anastomosis:** if tip reaches another vessel, vessels connect

4. **Maturation:** new vessel positions gradually increase `vascular_density` over `vessel_maturation_hr` hours
 5. **Effect:** new vascular positions become O_2 /glucose sources
-

6. ecDNA as Somatic Instrumental Variable (SIV)

6.1 The Causal Inference Problem

In observational spatial data, associations between molecular features (e.g., gene expression) and cellular phenotypes (e.g., proliferation) are confounded by shared microenvironment. A cell near a vessel has both higher O_2 *and* different gene expression; separating cause from correlation is impossible without randomization.

6.2 ecDNA Segregation as Natural Randomization

ecDNA elements segregate during mitosis via a mechanism that is:

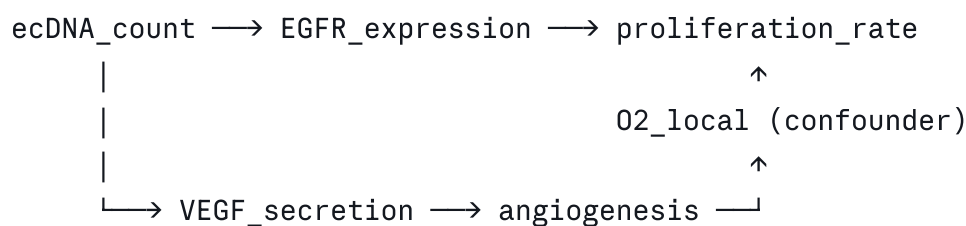
- **Random:** Each ecDNA copy is independently assigned to a daughter cell with probability $p \approx 0.5$ (binomial process)
- **Heritable:** Daughter cells inherit the randomly assigned count, which then influences their phenotype
- **Independent of confounders:** The physical segregation mechanism is not driven by the local microenvironment

This satisfies the three instrumental variable assumptions:

1. **Relevance:** ecDNA copy number affects the instrumented variable (e.g., EGFR expression)
2. **Independence:** ecDNA segregation is independent of confounders (microenvironment)
3. **Exclusion restriction:** ecDNA affects outcome only through the instrumented pathway

6.3 Ground-Truth Causal Graph

CAUSANTA embeds a known Structural Causal Model:



The simulation knows the true causal effect sizes ($\alpha, \beta, \gamma, \delta$ from Section 5.1). Causal discovery algorithms operating on the output spatial data should recover this structure. The degree to which they succeed — measured against these known ground-truth edges — constitutes the benchmark.

6.4 Counterfactual Trajectories

CAUSANTA supports **intervention experiments**: at any time point, the user can modify ecDNA status at specific positions and re-run the simulation forward to observe how the tissue evolves differently. Comparing the factual trajectory (observed ecDNA distribution) against counterfactual trajectories (modified ecDNA) quantifies causal effects spatially.

7. Initialization

7.1 Synthetic Initialization (Default Mode)

Normal brain tissue is populated according to biologically realistic proportions for cortical gray matter:

Cell Type	Fraction	Placement Rule
Neuron	40-50%	Distributed across parenchyma
Astrocyte	20-30%	Tiled in non-overlapping ~50-100 μm domains
Oligodendrocyte	10-15%	Interspersed, denser near white matter tracts
Microglia	5-10%	Scattered uniformly; 5-10% of all glial cells
Endothelial	~3-5%	Organized as capillary network segments
Pericyte	~1-2%	Adjacent to endothelial cells

Vascular network: Placed first as a branching capillary pattern with inter-capillary distance of ~100-200 μm . Pericytes assigned to positions adjacent to endothelial cells.

Tumor seeding: One or more tumor cells placed at specified coordinates with initial ecDNA_count, ecDNA_cargo, and cell cycle phase.

Environment equilibration: After placing cells and vasculature, the diffusion solver runs for a burn-in period (default 100 hours equivalent) to reach steady-state O_2 /glucose fields before the first recorded time step.

7.2 Data-Driven Initialization (for Real Data)

Load real spatial data and register to the CAUSANTA coordinate grid:

- **H&E:** Tissue architecture → ECM_density map, cell segmentation → positions and morphology
- **FISH:** Per-cell ecDNA copy number → `ecDNA_count` per cell
- **PhenoCycler / CODEX:** Protein panel → cell type classification, positions, protein expression
- **Visium:** 55 μm spots deconvolved to cell neighborhoods → transcriptomic features per cell
- **AlphaCycler:** Additional spatial proteomic channels

Environment fields initialized from tissue context (vessel positions from CD31 staining, ECM from collagen markers) rather than synthetic baseline.

8. Simulation Main Loop

INITIALIZE:

1. Load parameters from JSON configuration file
2. Create environment grid at specified resolution
3. Initialize environment fields to baseline values
4. Place vascular network (synthetic or from data)
5. Populate with normal cell types at specified proportions
6. Seed tumor cell(s) at specified position(s)
7. Equilibrate environment fields (burn-in diffusion)
8. Write initial state: `cells_t000000.tsv`, `environment_t000000.tsv`
9. Copy parameter JSON to output directory

FOR each time step $t = 1, 2, \dots, T_{\text{total}}$:

PHASE 1: ENVIRONMENT DIFFUSION

For each diffusion sub-step within the 1-hour main step:

For each substrate (O_2 , glucose, VEGF, lactate):

- a. Compute source terms (vascular supply, cell secretion)
- b. Compute sink terms (cell consumption)
- c. Solve diffusion via implicit LOD (Thomas algorithm)
- d. Apply natural decay (λ_k)
- e. Enforce boundary conditions and physical range clamps

Compute derived fields (pH from lactate, hypoxia flags)

PHASE 2: CELL BEHAVIOR (cells processed in random-shuffled order)

For each living cell *c*:

- a. Sample local environment via bilinear interpolation
- b. Update *is_hypoxic*, *is_quiescent* flags
- c. Death check:
 - Necrosis: O_2 below threshold for required duration?
 - Apoptosis: stochastic roll
- d. State transition check (Table 5.4)
- e. Proliferation:
 - Advance cell cycle clock
 - If M-phase complete and space available:
Execute division, partition ecDNA binomially
Create daughter with new *cell_id*, *parent_id*
- f. Migration:
 - Compute direction (gradients + persistence + noise)
 - Compute displacement (speed × 1 hr)
 - Attempt move (volume exclusion check)

PHASE 3: ANGIOGENESIS

For each endothelial cell with adjacent VEGF > threshold:

Attempt sprouting (Section 5.5)

PHASE 4: IMMUNE RECRUITMENT

For each vascular position with chemokine > threshold:

Stochastic spawn of recruited immune cell

PHASE 5: CLEANUP

Stochastic lysis of necrotic cells

Update environment grid occupancy

PHASE 6: OUTPUT (at configured interval)

Write *cells_t{step:06d}.tsv*

Write *environment_t{step:06d}.tsv*

Write summary statistics to log

9. Output Specification

9.1 Directory Structure

Each simulation run produces a timestamped output directory:

```
output/
  run_20260410_143022/
    params.json          # Complete parameter file (frozen copy)
    cells_t0000000.tsv   # Initial cell state
    cells_t0000001.tsv   # t = 1 hour
    cells_t0000002.tsv   # t = 2 hours
    ...
    environment_t0000000.tsv # Initial environment
    environment_t0000001.tsv
    ...
    summary.log          # Per-step statistics (cell counts, mean  $O_2$ ,
etc.)                    # etc.)
    lineage.tsv          # Complete parent-child lineage table
```

9.2 Cells File Format (TSV)

Filename: `cells_t{step:06d}.tsv`

Column	Type	Description
cell_id	int	Unique cell identifier
parent_id	int	Parent cell ID (-1 for initial cells)
cell_type	int	Type code (0-8)
cell_type_name	string	Human-readable type
x	int	X position in μm
y	int	Y position in μm
angle	float	Orientation (radians)
shape_path	string	SVG path string
size_scale	float	Scale factor for shape rendering
cell_cycle_phase	string	G0, G1, S, G2, or M
ecDNA_count	int	ecDNA copy number
ecDNA_cargo	string	Semicolon-delimited gene list
O2_local	float	Local oxygen (mmHg)
glucose_local	float	Local glucose (mM)
is_hypoxic	bool	True/False
is_quiescent	bool	True/False
generation	int	Divisions from ancestor
time_born_hr	float	Hour when cell was created
migration_rate	float	Current effective speed ($\mu\text{m/hr}$)
VEGF_secretion	float	Current VEGF output (amol/hr)

9.3 Environment File Format (TSV)

Filename: `environment_t{step:06d}.tsv`

Column	Type	Description
x_grid	int	Grid column index
y_grid	int	Grid row index
x_um	float	X in μm
y_um	float	Y in μm
O2	float	Oxygen (mmHg)
glucose	float	Glucose (mM)
VEGF	float	VEGF (nM)
lactate	float	Lactate (mM)
pH	float	Derived pH
ECM_density	float	ECM [0-1]
vascular_density	float	Vessel density [0-1]
is_occupied	bool	Cell present?
occupant_cell_id	int	cell_id or -1
occupant_type	int	Cell type or -1

9.4 Lineage File

Filename: `lineage.tsv`

Append-only file recording every division event:

Column	Type	Description
time_hr	float	Hour of division
parent_id	int	Dividing cell's ID
parent_ecDNA_before	int	Parent ecDNA count before division
parent_ecDNA_after	int	Parent ecDNA count after segregation
daughter_id	int	New daughter cell's ID
daughter_ecDNA	int	Daughter ecDNA count
x	int	Position of division (μm)
y	int	Position of division (μm)
generation	int	Generation number of daughter

10. Visualization (Vega.js)

10.1 Rendering Pipeline

The cells TSV is loaded directly into a Vega specification where each cell is rendered as a `path` mark:

1. Read `shape_path` as the Vega path datum
2. Apply transform: scale by $\text{size_scale} \times (\text{diameter}/2)$, rotate by `angle`, translate to (x, y)
3. Color by `cell_type` using the type color palette
4. Opacity modulated by state (e.g., necrotic cells at 0.3 opacity, hypoxic at 0.7)

10.2 Environment Overlay

The environment TSV renders as a `rect` grid heatmap behind the cell layer:

- Channel selectable: O_2 , glucose, VEGF, lactate, pH, ECM, vascular density
- Color scale: sequential (e.g., viridis for O_2 , magma for VEGF)
- Opacity adjustable to see cells through environment

10.3 Interactive Features

- Time slider: scrub through simulation steps
- Cell selection: click cell to see full state vector, lineage tree
- Layer toggles: show/hide cell types, environment channels
- Zoom: pan and zoom across the tissue section
- ecDNA highlight mode: cells colored by ecDNA_count (continuous scale)

11. Core Mathematical Relationships

Process	Equation	Key Parameters
Diffusion	$\partial E/\partial t = D \nabla^2 E - \lambda E + S$	D, λ per substrate
O ₂ vascular supply	$S = q \times V_{\text{asc}} \times (O_{2_blood} - O_2)$	q (transfer coeff), $O_{2_blood} = 60$ mmHg
O ₂ -dependent proliferation	$r = r_max \times \max(0, (O_2 - \theta_H)/(O_{2_max} - \theta_H))$	r_max, θ_H
ecDNA segregation	$daughter_ecDNA \sim \text{Binom}(N, p)$	$N = \text{parent count}, p = 0.5$
ecDNA → division	$T_eff = T_base / (1 + \alpha \log_2(1 + ecDNA))$	α (effect size)
ecDNA → VEGF	$S_eff = S_base \times (1 + \beta \times ecDNA)$	β
ecDNA → apoptosis resistance	$a_eff = a_base / (1 + \gamma \times ecDNA)$	γ
ecDNA → migration	$v_eff = v_base \times (1 + \delta \times ecDNA)$	δ
VEGF secretion	$S_VEGF = r \times H(\theta_hyp - O_2)$	r, θ_hyp
Migration	$v = v_persist + \chi \nabla O_2 + \chi_ECM \nabla ECM + \text{noise}$	χ coefficients
Necrosis	$\text{type} \rightarrow \text{Necrotic if } O_2 < \theta_nec \text{ for } > T_nec \text{ hours}$	θ_nec, T_nec
Immune kill	$P(\text{kill}) = k \times \text{activation} \times \Delta t$	k (kill rate)
pH from lactate	$pH = 7.4 - 0.02 \times [\text{lactate}]$	empirical approximation

12. Validation Targets

The simulation should reproduce these known biological phenomena:

1. **Avascular spheroid zonation:** Proliferating rim (~100–200 μm from vessels), quiescent intermediate zone, necrotic core — matching in vitro spheroid data
2. **Oxygen gradient:** Hypoxia onset at ~100–150 μm from nearest vessel
3. **Pseudopalisading necrosis:** Migrating waves of hypoxic cells around necrotic foci (GBM pathognomonic feature)
4. **ecDNA heterogeneity dynamics:** Inter-cell variance in ecDNA copy number increases with division count, following $\text{Var} = N \times p \times (1-p)$ per generation under binomial segregation
5. **Angiogenic switch:** VEGF accumulation triggers vessel sprouting as tumor mass exceeds ~1–2 mm
6. **Immune exclusion:** High tumor density + immunosuppressive signaling creates immune-cold core with immune cells restricted to tumor periphery
7. **Gompertzian growth:** Tumor growth curve transitions from exponential $\rightarrow$ linear $\rightarrow$ plateau as carrying capacity limits and necrosis increase
8. **Causal recoverability:** Causal discovery algorithms applied to output data should recover the ground-truth ecDNA $\rightarrow$ phenotype causal edges (Section 6.3) at rates significantly above chance

13. Implementation Architecture

```
causanta/  
  __init__.py  
  config.py           # JSON parameter loading and validation  
  domain.py           # Grid, coordinate system, boundary conditions  
  cells.py            # Cell agent class, state vector, type parameters  
  environment.py      # Environment fields, diffusion solver (LOD/Thomas)  
  behaviors.py        # Proliferation, migration, death, transitions  
  angiogenesis.py     # Vascular sprouting logic  
  ecdna.py            # ecDNA segregation, modulation effects  
  initialization.py   # Synthetic and data-driven tissue setup  
  simulation.py       # Main loop orchestration
```


io.py	# TSV/JSON reading and writing
visualization.py	# Vega.js spec generation
README.md	# Project documentation
memory.md	# Development log and decisions
params/	
default.json	# Default parameter configuration
output/	# Simulation run outputs (timestamped subdirs)

Performance strategy: Core simulation loop in Python (NumPy for environment field operations). If profiling reveals bottlenecks, migrate diffusion solver and cell iteration to Cython, Rust (via PyO3), or C extensions. The 100×100 environment grid with ~5,000–50,000 cells is well within pure-Python/NumPy performance for exploratory work.